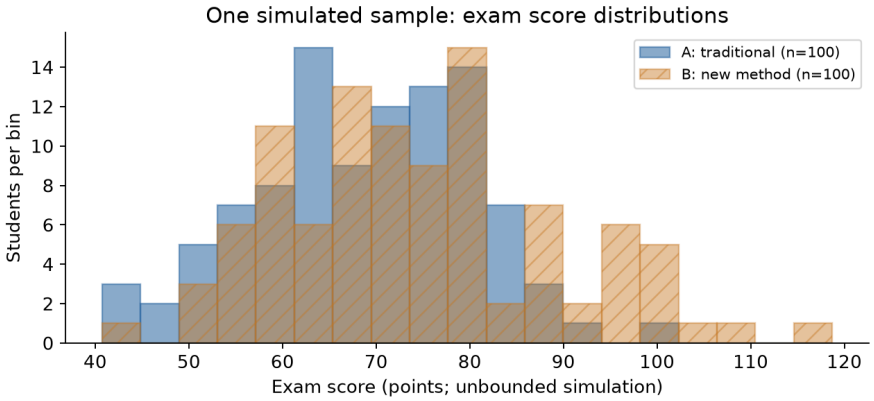
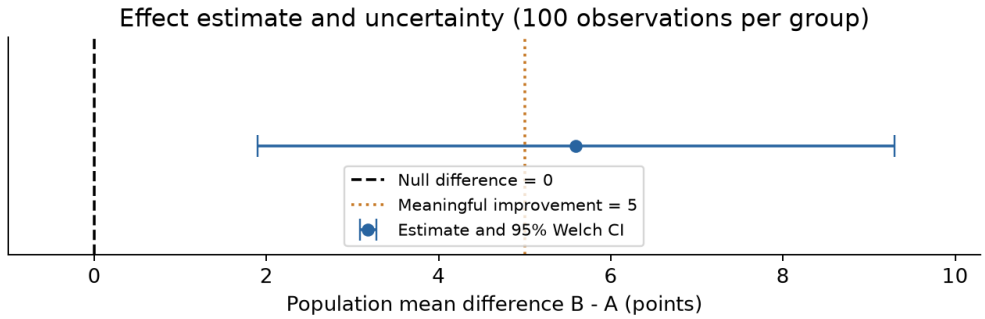
Question, sampling and inference

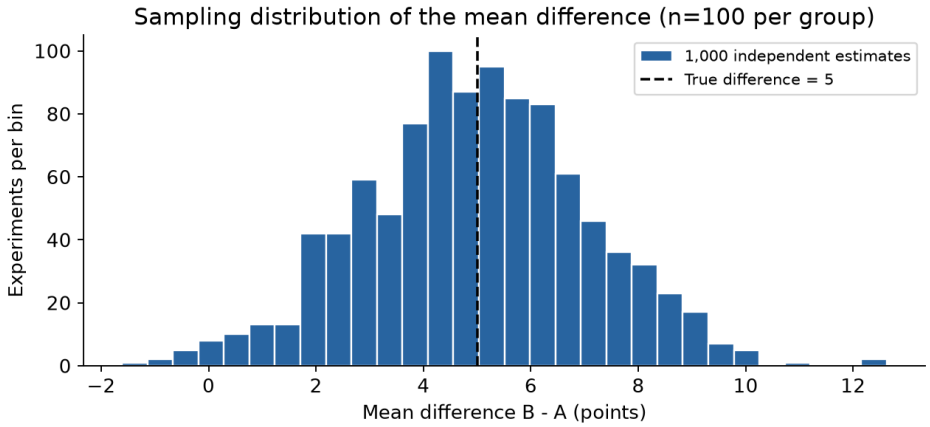
Simulated teaching scores: A mean=70, B mean=75, both SD=15, n=100 per group. Estimand: mean B minus mean A. H0: difference=0; H1: nonzero; alpha=0.05; meaningful improvement=5 points. Assume independent normal observations. Seeds: 42-45.

Observed means: A=69.246, B=74.841. Welch difference=5.595; 95% CI [1.896, 9.293]; p=0.003226; df=188.219. The CI extends below 5, so an improvement of at least 5 points is not established.





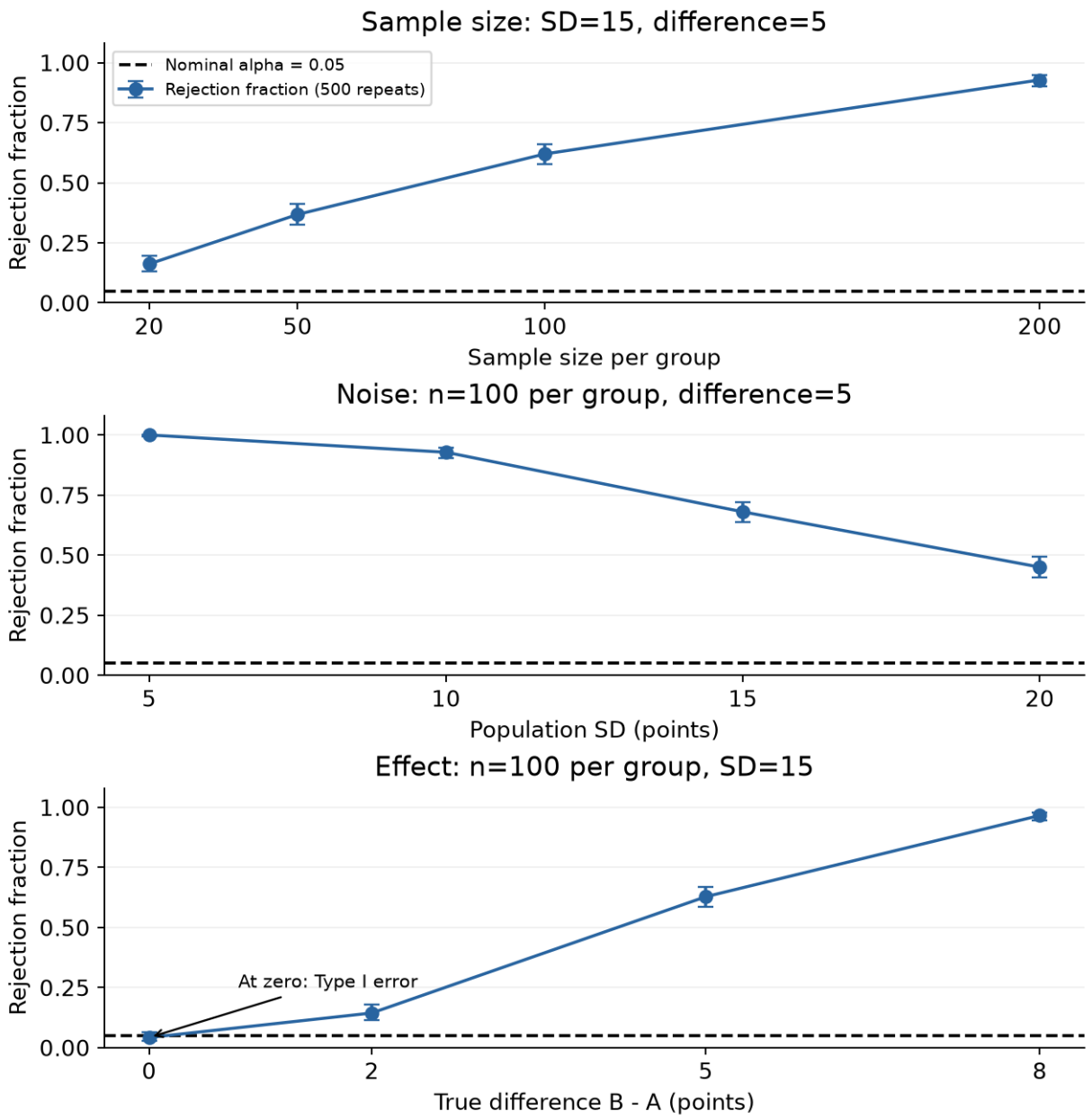
Welch uses SE=sqrt(sA squared/nA+sB squared/nB) and CI=difference +/- t(0.975, df) x SE. A 95% procedure covers the true difference in about 95% of repeated samples; p is not the probability that H0 is true.



In 1,000 samples, the estimate averaged 4.979, with SD 2.082; theoretical SE=2.121. Larger samples reduce sampling variation.

Statistical power

Each condition uses 500 Welch tests at alpha=0.05. Change one factor from n=100 per group, SD=15 and difference=5. Error bars are 95% Wilson intervals for the rejection probability.

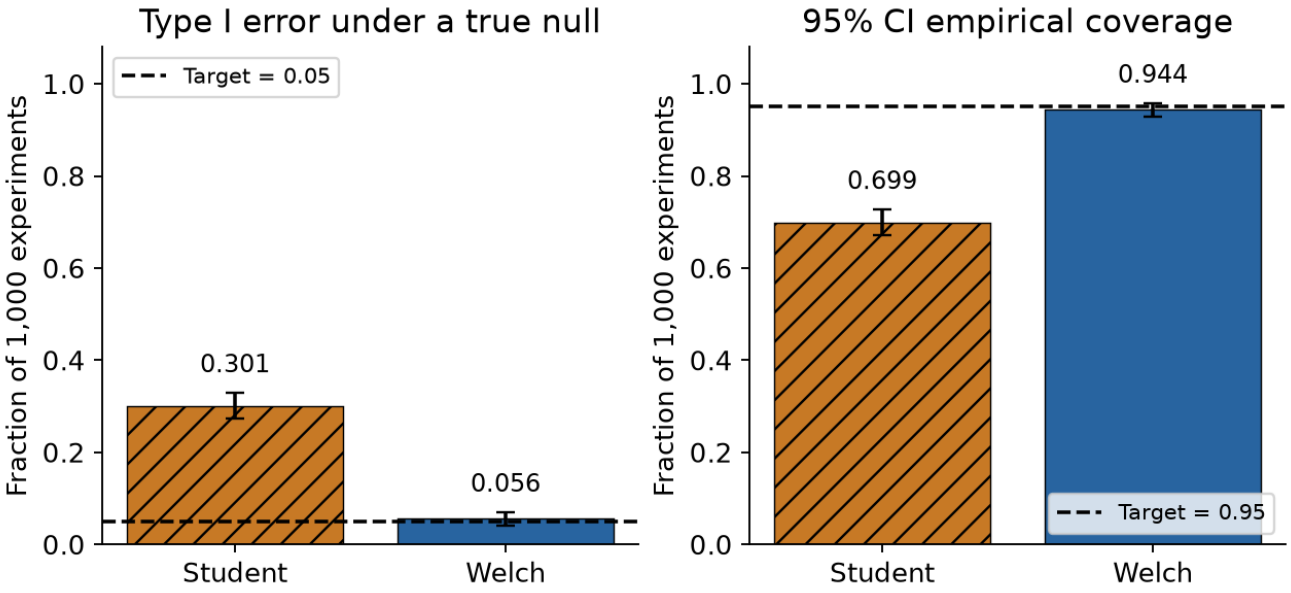


n=20/50/100/200 power: 16.2% / 36.8% / 62.0% / 92.8%.
SD=5/10/15/20 power: 100.0% / 92.8% / 68.0% / 45.0%.
Difference=0/2/5/8 rejection: 4.2% / 14.4% / 62.8% / 96.6%.

More observations, less noise and larger effects improve detection. At zero effect, rejection measures Type I error. Finite repetitions explain small differences between repeated baseline conditions.

Counterexample and limitations

Both means=70. A: n=25, SD=20; B: n=100, SD=5. Compare Student and Welch on the same samples in 1,000 experiments. This violates Student's equal-variance assumption, while independence and normality remain.



Student/Welch false positives: 30.1% / 5.6%; 95% CI coverage: 69.9% / 94.4%. Pooling overweights the large low-variance group and understates uncertainty from the small noisy group. Welch handles the two variances separately and reduces the problem here.

Checks: direct formulas agree with SciPy; saved CSVs match computed values; seeds make the results reproducible. The notebook includes the SE, degrees-of-freedom and CI calculations.

Limitations: scores are unbounded normal draws, observations are independent, and rejection/coverage estimates have Monte Carlo error. These simulations cannot establish a real teaching benefit.

Reference: SciPy, scipy.stats.ttest_ind. https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html